Assessment Task 3
[Visual Analytics]

32146 Data Visualisation and Visual Analytics – Autumn 2025

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Executive Summary

This report explores Australia's energy trade, focusing on petroleum as the dominant import category within Mineral Fuels, Lubricants, and Related Materials. Petroleum accounts for over 97% of Australia's energy imports in 2023, highlighting the country's reliance on external sources. Major events such as the Global Financial Crisis (2008), the Oil Price Crash (2016), the COVID-19 Pandemic (2020), and the Russia-Ukraine War (2022) have significantly impacted trade patterns, causing volatility in petroleum imports.

The analysis identifies that coal and gas dominate Australia's energy exports, while petroleum imports show sharp spikes during global crises. Forecasts suggest that petroleum imports will remain high but may gradually decline post-2024, with uncertainties driven by geopolitical and economic factors.

The report recommends that Australia diversify its energy sources, strengthen domestic supply chains, and explore renewable alternatives to reduce dependence on petroleum and enhance trade resilience.

1. Introduction

Australia's energy trade plays a crucial role in the nation's economy, with a strong reliance on the **Mineral Fuels, Lubricants, and Related Materials** category. Within this, **petroleum** stands out as the dominant import product, accounting for over **97% of Australia's energy imports in 2023**. This significant dependency has been shaped by global events, including the **Global Financial Crisis (2008)** and the **Russia-Ukraine War (2022)**, which led to substantial import spikes in recent years.

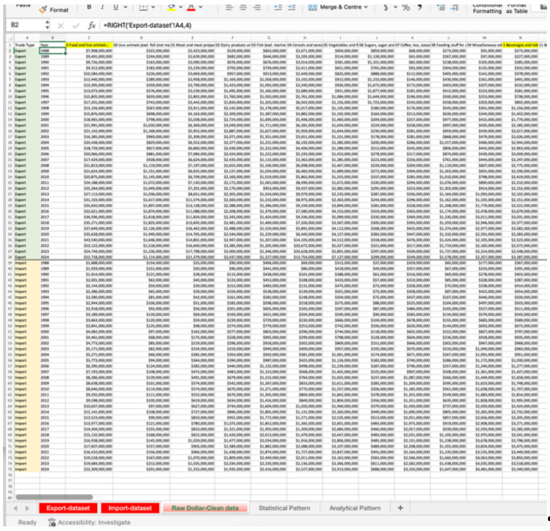
This report aims to

- **Analyse** Australia's energy trade patterns by examining the relationship between the **main category** (Mineral Fuels) and its **subcategories** (Coal, Petroleum, and Gas).
- **Visualise** petroleum import trends, highlighting key **trends**, **volatility**, and **outliers** observed between 2015 and 2024.
- **Forecast** potential risks and opportunities for Australia's petroleum imports over the next five years (2025–2029).
- Provide **actionable recommendations** to strengthen Australia's energy import strategy while reflecting on the strengths and limitations of the data visualisation methods used.

2. Data Preparation and Visualisation Process

2.1 Data Source and Cleaning

- **Create 3 Excel sheet (Raw Dollar, Statistical Pattern, Analytical Pattern)**
- **Raw Dollar-Clearn data**



<- Extract data source from Export-dataset and Import-dataset

- **Statistical Pattern;** showing how much contribute to the Total (Ratio)

  – Show how much each main category contributes to the total, helping identify which items are the main exports/imports.
  – Data source: *Raw Dollar - Clean Data*.
  – Calculate the ratio of main categories relative to the total, and subcategories relative to each main category.

- **Analytical pattern;** showing how much increased or decreased compared to the previous year

  – Calculate how much each category has increased or decreased compared to the previous year.
  – Note: I could not calculate the change for 1988 due to missing previous-year data.
  – Data source: *Raw Dollar - Clean Data*

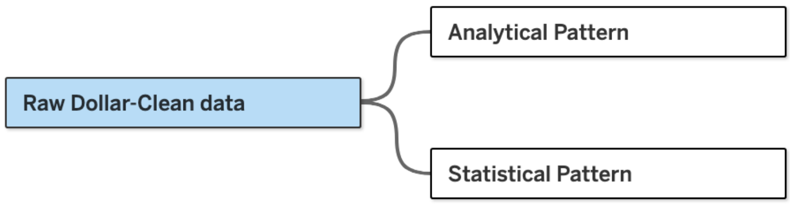
Therefore, we deleted the row"1988 year" in export & import

Trade Type	Year	0 Food and li	00 Live anima	01 Meat and	02 Dairy prod	03 Fish
Export	1989	120.02%	75.78%	108.91%	128.92%	9
Export	1990	102.48%	66.80%	117.09%	99.12%	10
Export	1991	95.74%	111.66%	104.82%	117.31%	10

37	Export	2024	96.29%	90.13%
38	Import	1989	114.28%	98.05%
39	Import	1990	99.22%	80.13%
40	Import	1991	104.55%	51.24%
41	Import	1992	107.15%	95.16%

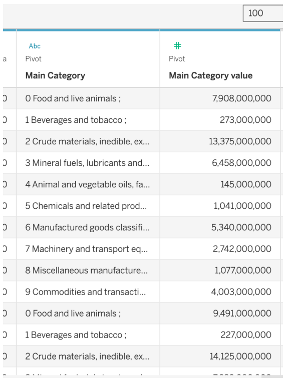
2.2 Import into Tableau

Raw Dollar-Clean data+ (Ass3 draft)



I used three data sources and created the fields "Main Category" and "Main Category Value" in each of them by pivoting the data field that includes all the main categories.

*** Result**



Pivot Main Category	Pivot Main Category value
0 Food and live animals ;	7,908,000,000
1 Beverages and tobacco ;	273,000,000
2 Crude materials, inedible, ex...	13,375,000,000
3 Mineral fuels, lubricants and...	6,458,000,000
4 Animal and vegetable oils, fa...	145,000,000
5 Chemicals and related prod...	1,041,000,000
6 Manufactured goods classifi...	5,340,000,000
7 Machinery and transport eq...	2,742,000,000
8 Miscellaneous manufacture...	1,077,000,000
9 Commodities and transacti...	4,003,000,000
0 Food and live animals ;	9,491,000,000
1 Beverages and tobacco ;	227,000,000
2 Crude materials, inedible, ex...	14,125,000,000

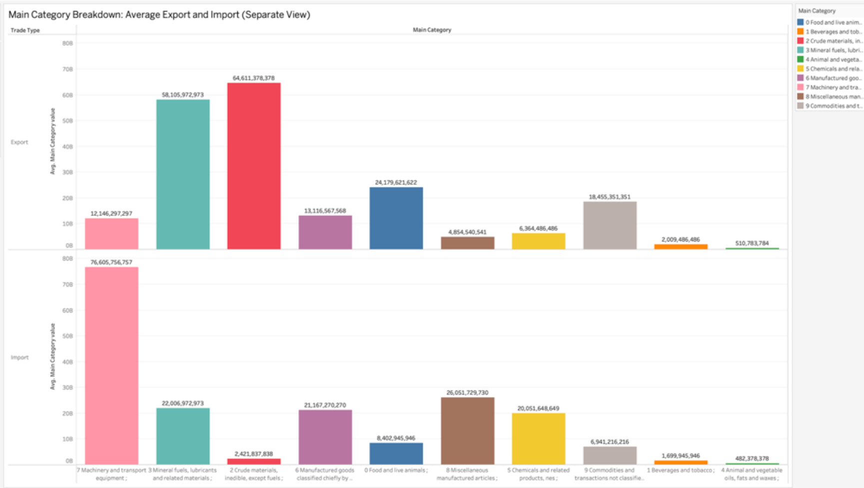
📊 Graphical Techniques You Used

- ☐ **Line Charts**
  (e.g., Petroleum Import Trends, Yearly Trends)
- ☐ **Bar Charts**
  (e.g., Yearly Import & Export Comparisons by Category)
- ☐ **Stacked Bar Charts**
  (e.g., Main Category Trends: Export & Import Stacked by Coal, Gas, Petroleum)
- ☐ **Pie Charts**
  (e.g., Import Composition Ratio - Petroleum, Coal, Gas)
- ☐ **Tree Maps (or Heatmaps)**
  (e.g., Petroleum Import Volatility by Year)
- ☐ **Forecast Line Charts (with confidence intervals)**
  (e.g., Petroleum Import Forecasts 2025–2029 with CI 95%)

3. Main Categories Analysis

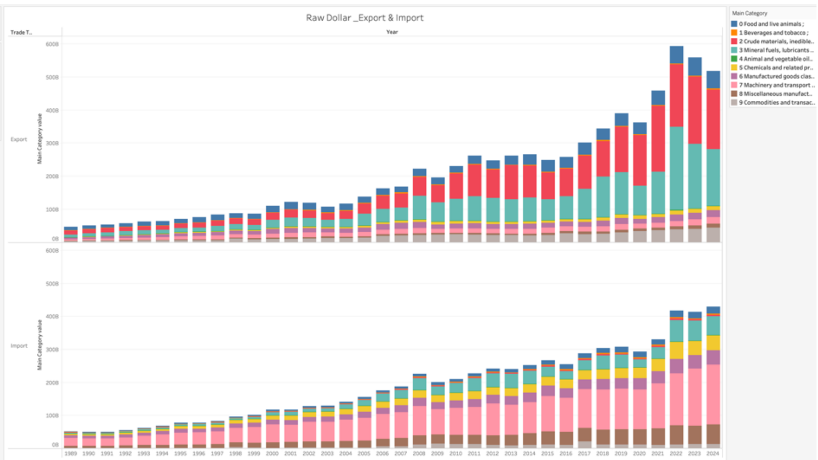
3.1 Raw Dollar value Trends

➤ All Main Categories Breakdown – Average Export and Import



<Figure 1: Main Category Breakdown – Average Export & Import>

⇨ This bar chart compares the average dollar values of Australia's export and import across main categories. It shows that **Machinery and Transport Equipment (Category 7)** is the leading import category, while **Mineral Fuels (Category 3)** dominates exports. The chart highlights the dependency on imported technology and the significance of energy resources in exports.



<Figure2: Yearly Trends in Raw Dollar Value – Export & Import>

⇨ This stacked bar chart shows **yearly Export and Import trends** across all categories from 1989 to 2024.

💡 Key insights

- Both Export and Import values have grown steadily, with a significant acceleration post-2000.
- Major external events such as **COVID-19 (2020)** and the **Russia-Ukraine war (2022)** are visible as dips and spikes in the trade patterns.
- The trade composition diversifies over time, with a broader range of categories contributing to total values after 2010.

3.2 Analytical Trends

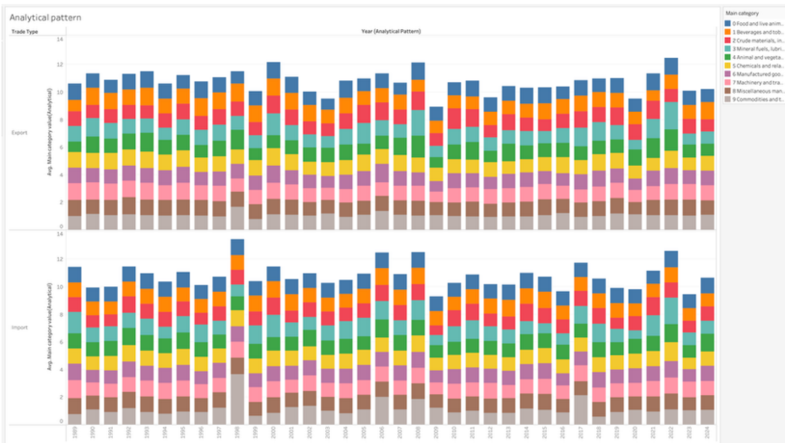
➤ Analytical Pattern Line chart

<Figure 3: Main Category Breakdown – Average Export & Import>

⇨ This line chart highlights **category-level fluctuations** across Export and Import from 1989–2024.

- **Mineral Fuels** and **Miscellaneous Manufactured Articles** exhibit the highest volatility.
- Sharp spikes in **1998, 2007, and 2022** correspond to global economic events (e.g., financial crises, energy shocks).
- Other categories remain stable, underscoring the impact of external factors on specific sectors.

➤ Main Categories Contribution to Export & Import by year



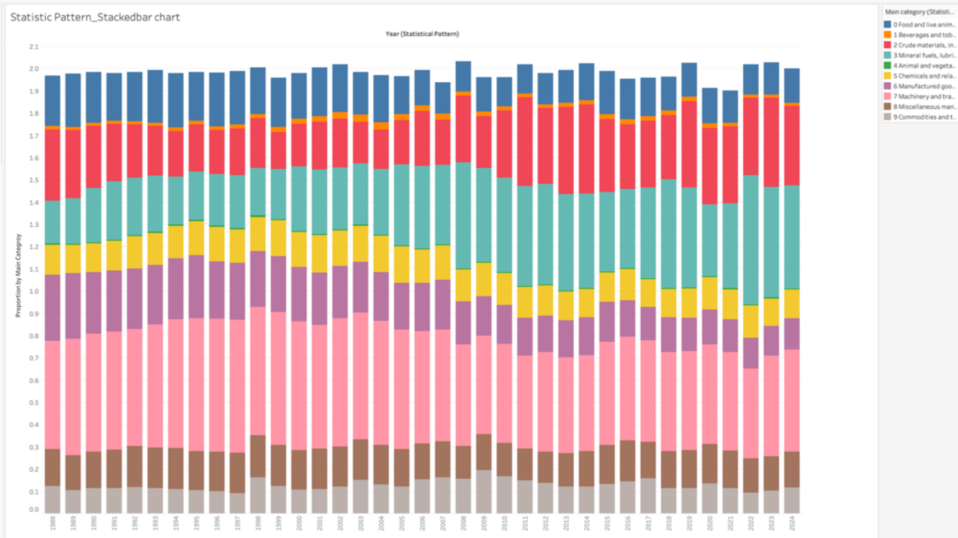
<Figure 5: Category Contribution to Export & Import Over Time (Stacked Bar View)>

⇨ This stacked bar chart shows the distribution of category contributions across Export and Import from 1989–2024.

- Both trade types display a **consistent composition**, with dominant categories (Mineral Fuels, Machinery, Crude Materials) maintaining a large share.
- Slight shifts in category proportions reflect evolving trade policies and global demand.

3.3 Statistical Trends

➤Visualizing Australia's Trade Structure: Stacked Bar Analysis

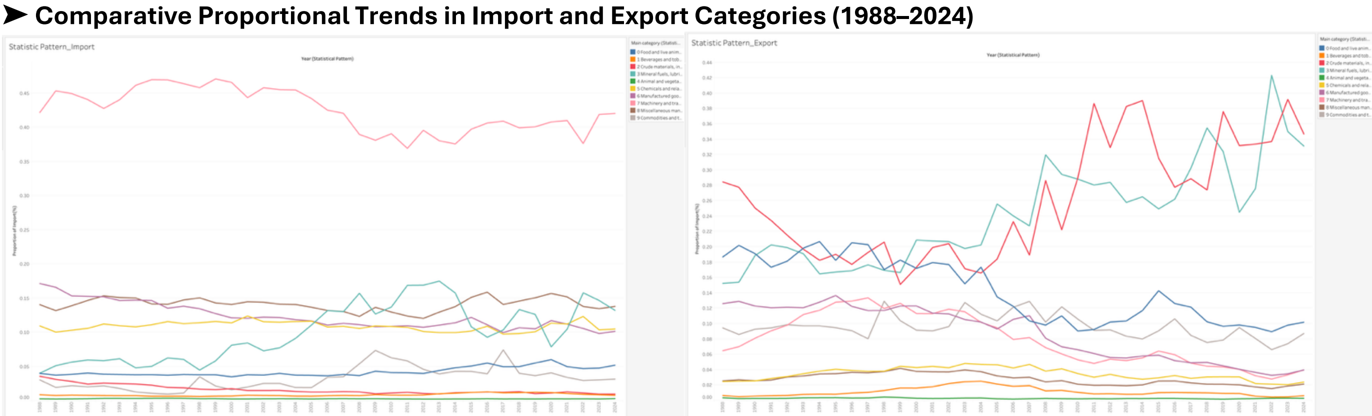


<*Figure 6:* Proportional Composition of Export and Import Categories Over Time (Stacked Bar View, 1988–2024)>

⇨ This stacked bar chart displays the **proportion of each trade category in total trade (combined Export and Import)** across years. It provides a **visual summary of how trade composition evolves over time**, highlighting category dominance, stability, and fluctuations.

- **Machinery and Transport Equipment (Category 7)** maintains a **consistently high proportion**, reflecting Australia's heavy import dependency on machinery and equipment.
- **Mineral Fuels (Category 3)** and **Crude Materials (Category 2)** show significant proportions in **exports**, contributing to overall trade value, but their shares fluctuate due to global commodity market shifts.
- **Food and Live Animals (Category 0)** holds a **stable share**, typically around 10–15%, representing Australia's steady agricultural exports.
- Categories like **Chemicals (5)**, **Miscellaneous Articles (8)**, and **Manufactured Goods (6)** exhibit relatively **small but stable proportions**, indicating consistent demand across sectors.
- Overall, the **relative composition remains stable**, with minor changes in some categories across different economic cycles and global events (e.g., commodity price spikes, trade policy shifts).

➤ Comparative Proportional Trends in Import and Export Categories (1988–2024)



<*Figure 7:* Proportional Trends in Import and Export Categories Over Time (1988–2024)>

⇨ This combined line chart visualizes the **proportion of each main category in Australia's total Import and Export trade** over the period 1988–2024. The focus is on identifying structural patterns, key differences, and trade dependencies based on category-level shares.

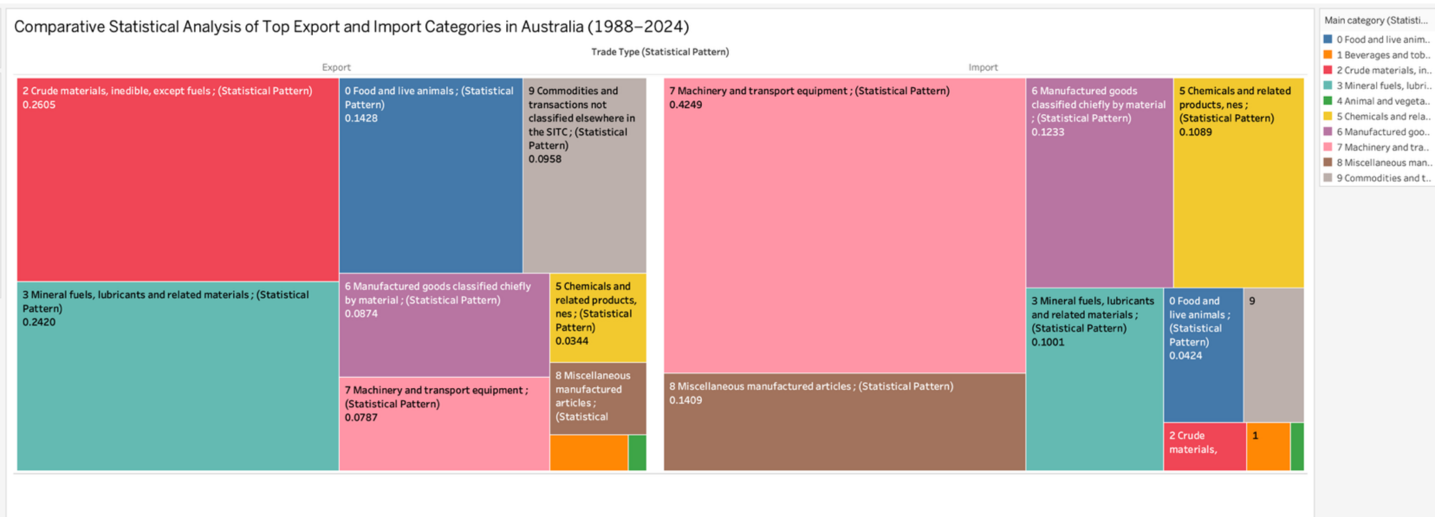
🏗️ **Import Trends**

- **Machinery and Transport Equipment (Category 7)** consistently dominates imports, maintaining a high share of **~40–45%** throughout the period, highlighting Australia's heavy reliance on imported technology and equipment.
- Other import categories like **Manufactured Goods (6)**, **Miscellaneous Manufactured Articles (8)**, and **Chemicals (5)** each contribute **~10–15%**, showing relatively stable but minor shares.
- **Mineral Fuels (3)** and **Crude Materials (2)** remain **low in import share (~10% or less)**, underscoring Australia's **self-sufficiency in energy**.

Export Trends

- **Crude Materials (2)** and **Mineral Fuels (3)** dominate exports, but their proportions exhibit **significant volatility**, with **Crude Materials** peaking at over **25%** in the 2000s and early 2020s before declining.
- **Food and Live Animals (0)** maintains a **stable 10–15% share**, demonstrating Australia's agricultural export base.
- Other categories (e.g., **Manufactured Goods (6)**, **Miscellaneous (8)**) remain marginal, reflecting a narrow export focus on raw resources rather than value-added goods.

➤ Proportional Distribution of Export and Import Categories (Treemap View)

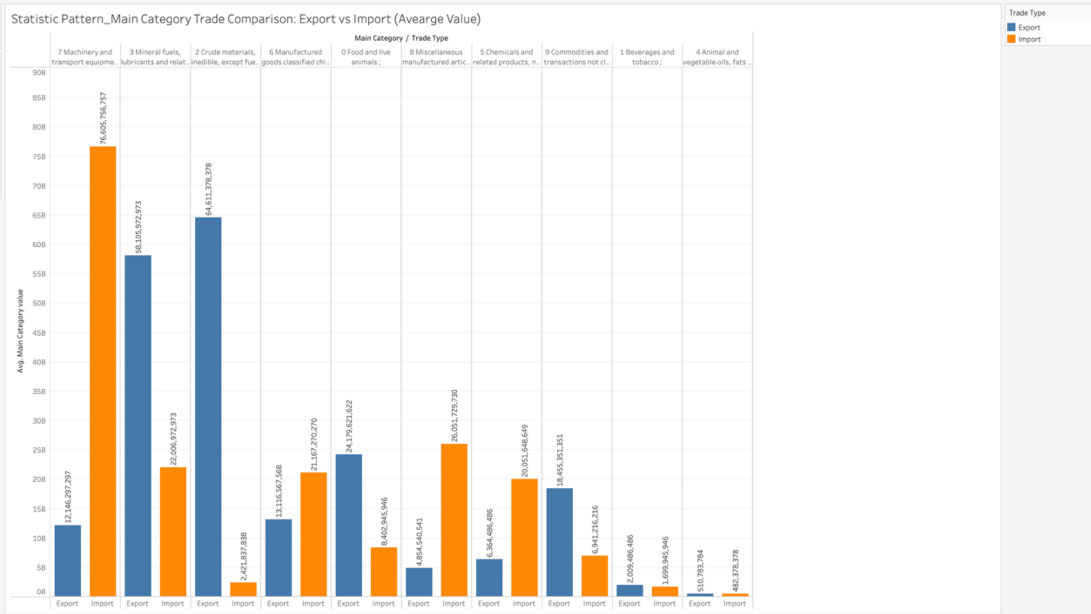


<***Figure 8:*** **Main Category Proportion by Export vs Import (Treemap View)**

⇨ This tree map visualizes the proportional breakdown of Australia's Export and Import by main category (1988–2024).

- Australia's **exports** are primarily driven by **raw materials** (Crude Materials + Mineral Fuels), while **imports** rely heavily on **manufactured goods and machinery**.
- The stark contrast—**42.5% Machinery in Imports vs. 7.8% in Exports**—highlights a structural dependency on foreign technology and equipment.
- Lower proportional shares for food, chemicals, and other consumer goods indicate a **narrow export focus** on natural resources.

➤ Average Trade Value Comparison by Main Category (Export vs Import)

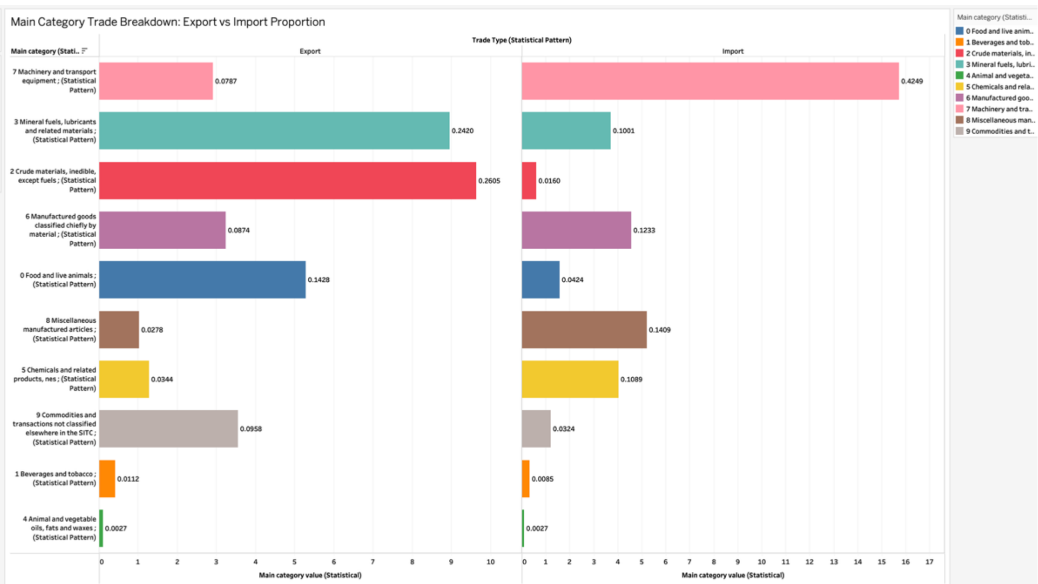


<Figure 9: **Main Category Proportion by Export vs Import (Treemap View)**

⇨ This bar chart compares the **average dollar values** of **Export** and **Import** across Australia's main trade categories from 1988 to 2024.
Key insights:

- **Machinery and Transport Equipment (Category 7)** dominates **Imports** with an average of **76.6 billion AUD**, reflecting a high reliance on imported machinery and technology.
- **Mineral Fuels (Category 3)** and **Crude Materials (Category 2)** are the top categories for **Exports**, with average values of **64.6 billion AUD** and **58.1 billion AUD**, respectively.
- Other categories, such as **Manufactured Goods (Category 6)** and **Miscellaneous Articles (Category 8)**, contribute modestly but consistently across both trade types.
- The chart clearly illustrates the structural trade imbalance: Australia primarily **imports high-value machinery** while **exporting raw materials and energy resources**.

➤ Identifying Trade Concentration and Imbalance: A Proportional View



<Figure 10: **Proportional Breakdown of Export vs. Import Categories (Statistical View, 1988–2024))**

⇨ This horizontal bar chart compares the **relative proportion of each trade category** in **exports and imports** over the period 1988–2024.

Exports (Left Panel)

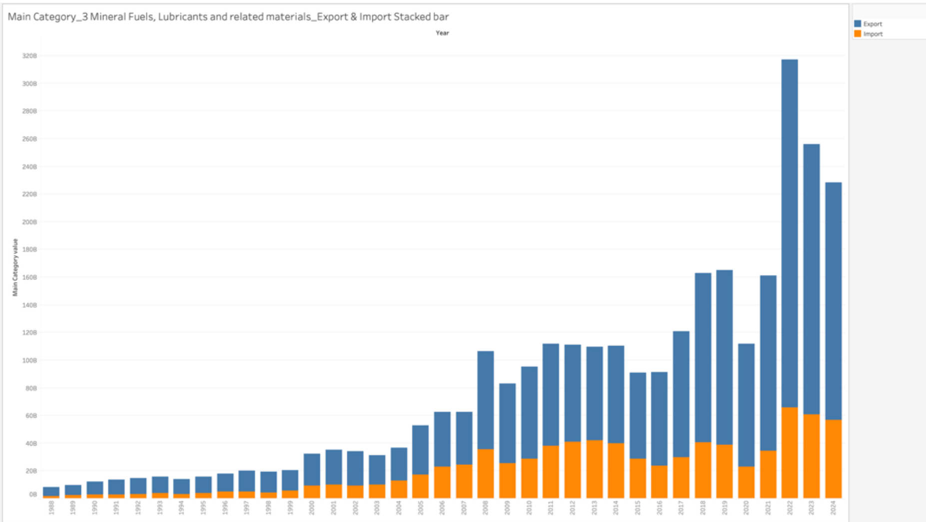
- **Crude Materials (2)**: Largest export category (26.05%), followed by **Mineral Fuels (3)** at 24.20%.
- **Food and Live Animals (0)** also plays a notable role at 14.28%.
- Other categories like **Machinery (7)** and **Miscellaneous Articles (8)** have smaller shares (<10%).

Imports (Right Panel)

- **Machinery and Transport Equipment (7)** overwhelmingly dominates at 42.49%.
- Other significant categories include **Miscellaneous Articles (8)** at 14.09% and **Manufactured Goods (6)** at 12.33%.
- Resource categories such as **Mineral Fuels (3)** (10.01%) and **Crude Materials (2)** (1.60%) are far less prominent in imports.

4. Main Category: Mineral Fuels, Lubricants and related materials

➤ Comparison and Growth Analysis of Mineral Fuels, Lubricants and related materials in Australia's Trade

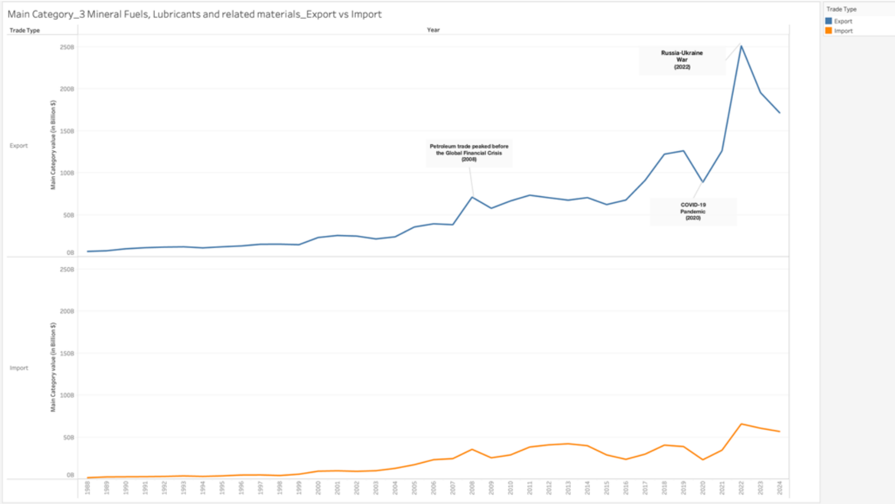


<Figure 11: **Export and Import Trends for Mineral Fuels, Lubricants, and Related Materials (1988–2024)>**

⇨ This stacked bar chart shows the **Export and Import values of Mineral Fuels (Category 3)** over time.

- **Exports** dominate the category, showing a **sharp upward trend from the early 2000s**, peaking dramatically after **2020**.
- **Major Change Points:**
  - **Early 2000s Surge:** Driven by increasing global demand, especially from China, and Australia's expanding LNG industry.
  - **2008-2009 Global Financial Crisis:** Slight dip visible in 2009 due to global economic slowdown.
  - **2020-2022 Spike:** Reflects energy price surges linked to **Russia-Ukraine War** and global supply shocks.
- **Imports** remain **minimal but steady**, indicating that Australia primarily produces mineral fuels for export and consumes a smaller proportion domestically.

➤ Mineral Fuels Trade Trends and External Factors: Export Surge, Geopolitical Impact, and Import Patterns



<Figure 12: Export and Import Trends of Mineral Fuels, Lubricants, and Related Materials (1988–2024)>

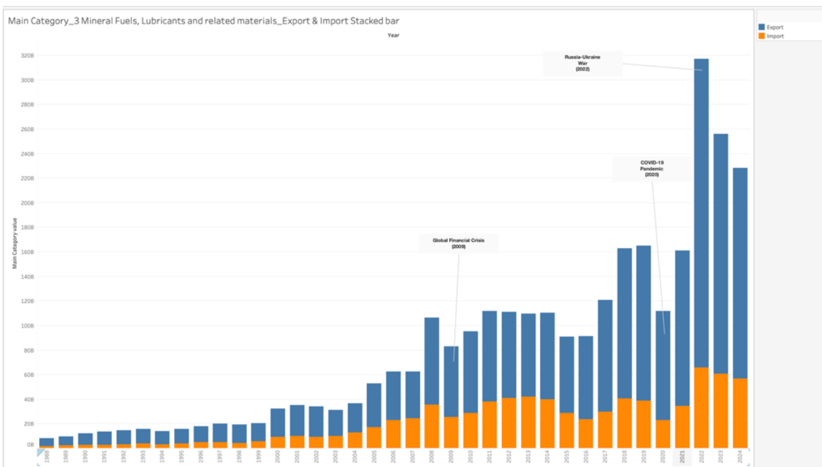
💡 Key Trends & Change Points

- **Exports:**
  - Steady growth from 1988 → sharp increases post-2000.
  - **2008 spike**: Petroleum trade peaked **before the Global Financial Crisis**.
  - **2020 dip**: COVID-19 pandemic → global demand shock, supply chain disruption.
  - **2022 surge**: Russia-Ukraine War → European demand for Australian energy skyrocketed **record-high exports**.
  - **2023-2024 correction**: Post-war stabilization and market adjustments, exports declined but still high.
- **Imports:**
  - Low baseline compared to exports, but gradual growth until **2012-2014** (oil price fluctuations).
  - Sharp increase in **2021-2022**: Linked to energy crisis (Russia-Ukraine War) and demand surge.
  - Slight decline in 2023 → market corrections.

🌍 External Factors Driving Trends

- **Global Financial Crisis (2008)**: Petroleum market instability.
- **COVID-19 (2020)**: Major disruption in trade flows, demand collapse.
- **Russia-Ukraine War (2022)**: **Energy supply shock**—Europe turning to Australia for alternative energy sources, boosting both export and import volumes.
- **Oil price fluctuations (2014–2016)**: Impact on trade values.

➤ Global Events Impacting Australia's Mineral Fuels Trade

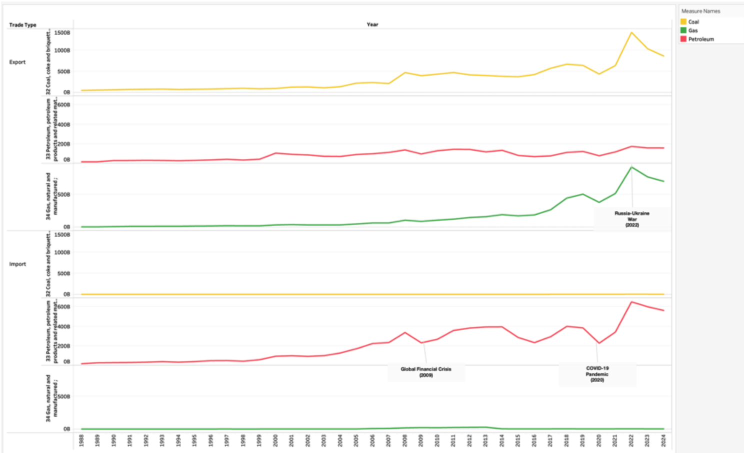


<Figure 13: Export & Import Trends of Mineral Fuels (1988–2024)>

💡 Key Insights

- **Exports**: Sharp increase, especially in **2022** due to the **Russia-Ukraine War** and global energy demand surge.
- **Imports**: Gradual rise, slight drop in **2020 (COVID-19)**, small uptick during **2022 energy crisis**.
- **Major Change Points**: 2009 Global Financial Crisis, 2020 COVID-19 Pandemic, 2022 Russia-Ukraine War.

➤ Subcategory Trends & Global Events Impact

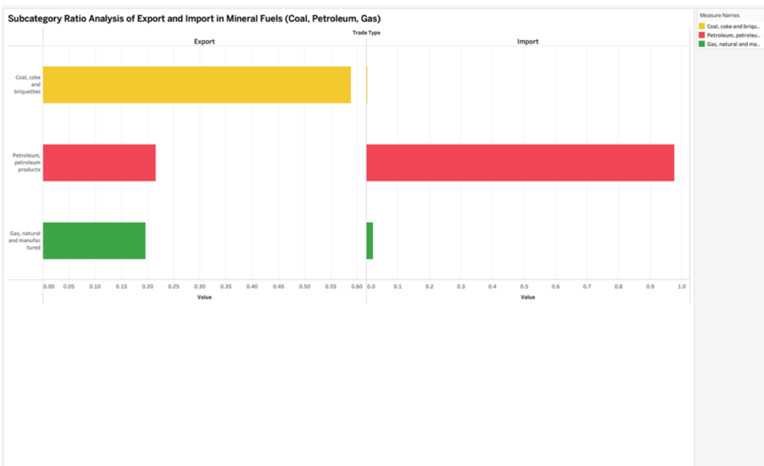


<Figure 14: Coal, Gas, and Petroleum Export & Import Trends (1988–2024)>

💡 Key Insights

- **Export Side**
  - **Coal**: Relatively stable growth, major surge around **2022** (Russia-Ukraine War), reflecting global energy crisis impact.
  - **Gas**: Gradual increase, peak around 2022; sensitive to global demand shocks, especially during energy market disruptions.
  - **Petroleum (Export)**: Fluctuations, but generally stable—less responsive compared to imports.
- **Import Side**
  - **Petroleum**: Dominates import trends; highly volatile. Peaks during **Global Financial Crisis (2008)**, **COVID-19 (2020)**, and sharp rise in **2022** (energy crisis from Russia-Ukraine War).
  - **Coal & Gas Imports**: Minimal change, very stable over time.
- **Overall**
  - **Petroleum** is the most volatile and sensitive to global crises.
  - **Coal & Gas Exports** show steady growth, strongly influenced by geopolitical events and energy demand surges.
  - Major change points: **2008 GFC**, **2020 Pandemic**, **2022 Energy Crisis**.

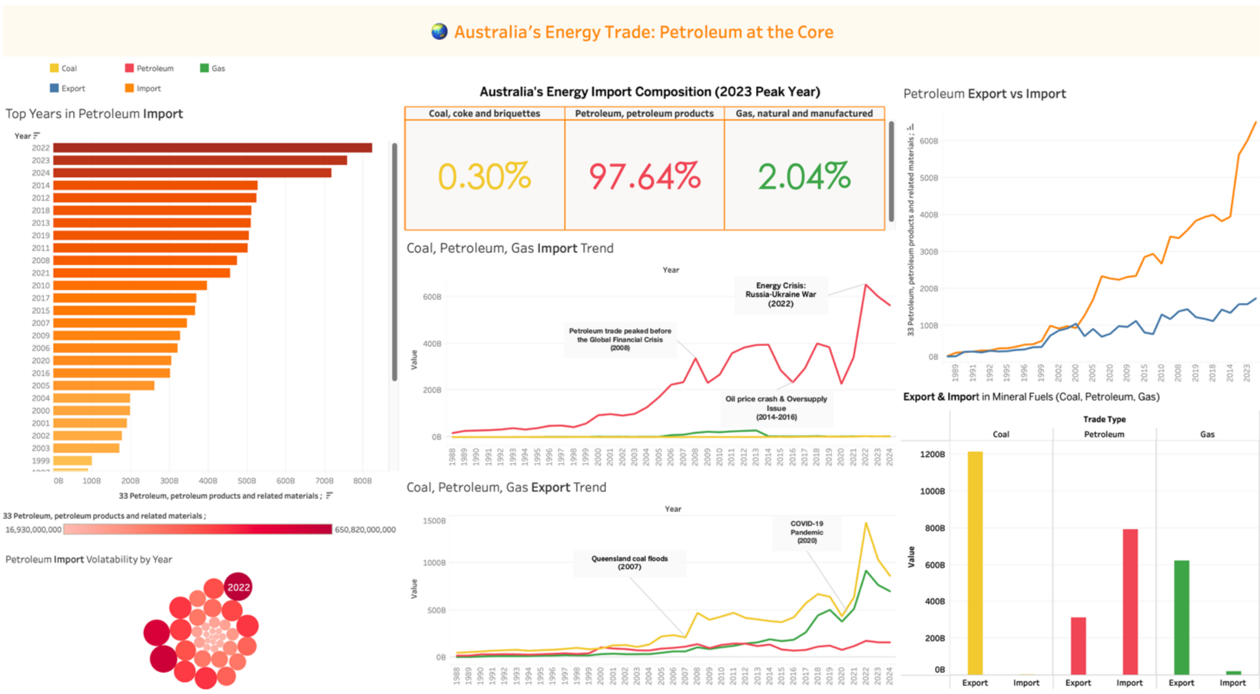
➤ Export vs Import Ratios in Mineral Fuels



💡 **Key Insights**

- **Export**:
  - **Coal dominates (≈60%)**, making it the primary export product.
  - **Petroleum and Gas** have much smaller shares (around 20% each).
  - **Export focus**: Australia mainly exports **Coal**, aligning with its natural resource abundance.
- **Import**:
  - **Petroleum dominates (≈90%)**: Australia heavily depends on petroleum imports due to limited refining capacity.
  - **Coal and Gas imports are minimal**, reinforcing Australia's position as a **net exporter** of these resources.
- **Overall**
  - **Coal is Australia's major export**, while **Petroleum is the key import driver** in the mineral fuels category.
  - Highlights Australia's **resource trade imbalance**: exporting raw coal, but importing refined petroleum products.

4. Dashboard with Petroleum Lead



<Figure 14: **Australia's Energy Trade: Petroleum at the Core)>**

📊 **Top Left: Top Years in Petroleum Import**

- **Purpose:** Shows the years with the highest petroleum import volumes.
- **Key Insights:**
  - **2022 and 2023** dominate, reflecting the **energy crisis post-Ukraine war**.
  - The rising trend since 2014 suggests **structural dependence** on imported petroleum.
  - **High volatility** post-2020 likely linked to **COVID-19 disruptions**, **oil price shocks**, and **geopolitical events**.
- **Your Insight Angle:** Use this to emphasize Australia's vulnerability to external shocks in petroleum markets.

📊 **Top Centre: Energy Import Composition (2023 Peak Year)**

- **Purpose:** A snapshot of import composition by main fuel types.
- **Key Insights:**

  - **Petroleum accounts for ~98%** of energy imports—this is **the core of your argument**.
  - Coal (0.3%) and gas (2%) barely register, showing **limited diversification**.
- **Your Insight Angle:** Argue that Australia's **import dependency is almost entirely petroleum-driven**, raising **energy security concerns**.

📊 **Top Right: Petroleum Export vs Import Trend**

- **Purpose:** Direct comparison of export vs import flows for petroleum.
- **Key Insights:**
  - Imports (orange) **skyrocket post-2000**, far outpacing exports (blue).
  - Key change points: **GFC (2008)**, **COVID-19 (2020)**, and **Russia-Ukraine war (2022)**.
- **Your Insight Angle:** Highlight Australia's **limited export capability** for petroleum despite heavy import reliance.

📊 **Middle Left: Petroleum Import Volatility by Year (Bubble Chart)**

- **Purpose:** Shows **volatility intensity** in petroleum imports over time.
- **Key Insights:**
  - **2022 has the largest volatility bubble**, aligning with the **energy crisis**.
  - Bubbles vary in size, reflecting **market shocks**.
- **Your Insight Angle:** Show that **import volatility peaks during global crises**, emphasizing **risk exposure**.

📊 **Middle Centre: Import Trend - Coal, Petroleum, Gas**

- **Purpose:** Time-series breakdown by fuel type.
- **Key Insights:**
  - **Petroleum imports dominate**, with **sharp spikes** during crises.
  - **Gas imports remain minimal**, coal imports almost **non-existent**.
- **Your Insight Angle:** Use this to show **structural dependence on petroleum imports**, especially in recent years.

📊 **Middle Right: Export Trend - Coal, Petroleum, Gas**

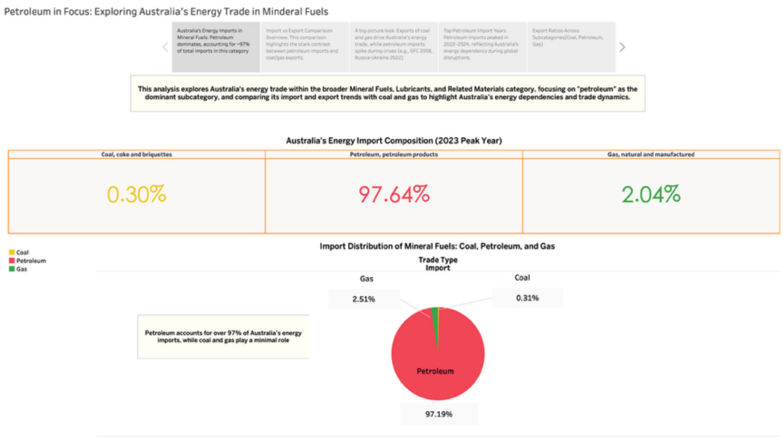
- **Purpose:** Time-series breakdown by fuel type for exports.
- **Key Insights:**
  - **Coal exports dominate** (e.g., post-2007 Queensland floods recovery).
  - **Gas exports grow** post-2010, reflecting **LNG boom**.
  - **Petroleum exports lag**—this contrast supports your focus.
- **Your Insight Angle:** Emphasize **Australia as a coal exporter but a petroleum importer**, revealing a **trade imbalance**.

📊 **Bottom Centre: Export & Import Comparison by Fuel Type (Bar Chart)**

- **Purpose:** Final comparative summary.
- **Key Insights:**
  - **Coal:** High export, negligible import.
  - **Petroleum:** High import, low export.
  - **Gas:** Moderate export, minimal import.
- **Your Insight Angle:** This chart visually confirms the **petroleum-centric trade imbalance**.

5. Storyboard Insights: Petroleum Trade Story

- **Slide 1: Import Composition Overview**

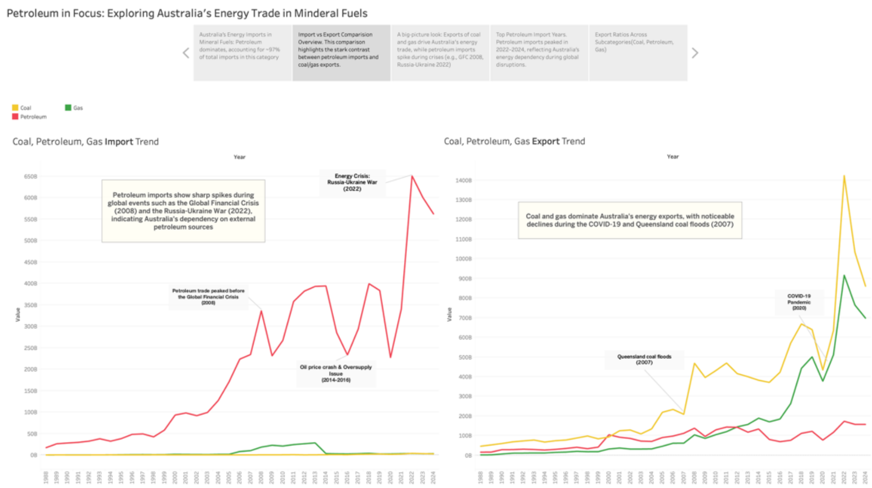


<Figure 15: Australia's Energy Import Composition (2023 Peak Year)>

– **Insight**: Petroleum overwhelmingly dominates Australia's energy imports, accounting for ~97%. Coal and gas are minimal contributors (<3% combined).

– **Key takeaway**: Australia relies heavily on petroleum for energy needs, creating a potential risk in energy security due to single-source dependency.

- **Slide 2: Import and Export Trends**

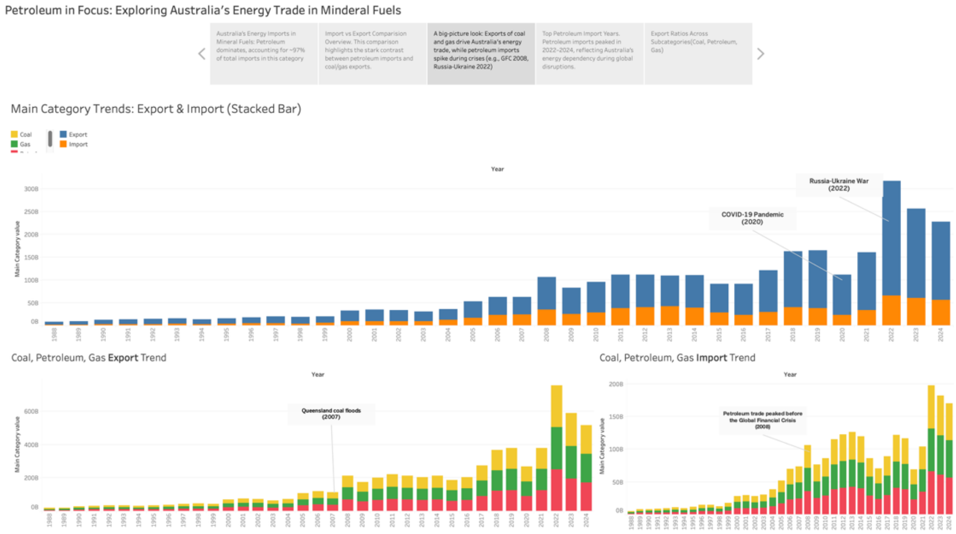


<Figure 16: Coal, Petroleum, Gas Import vs Export Trends>

– **Insight**
  - Petroleum imports show sharp volatility with global events (GFC 2008, Oil Price Crash 2014–16, Russia-Ukraine War 2022).
  - Coal and gas dominate exports with more stable patterns.
  - COVID-19 (2020) caused a noticeable dip in coal/gas exports, but petroleum imports remained high.

– **Key takeaway**: Australia's export strength lies in coal and gas, while petroleum imports expose vulnerability during global crises.

- **Slide 3: Main Category Trends: Export & Import (Stacked Bar)**



<*Figure 17:* **Australia's Energy Trade Trends – Export & Import Overview by Main Categories (Coal, Petroleum, Gas)*:* >**

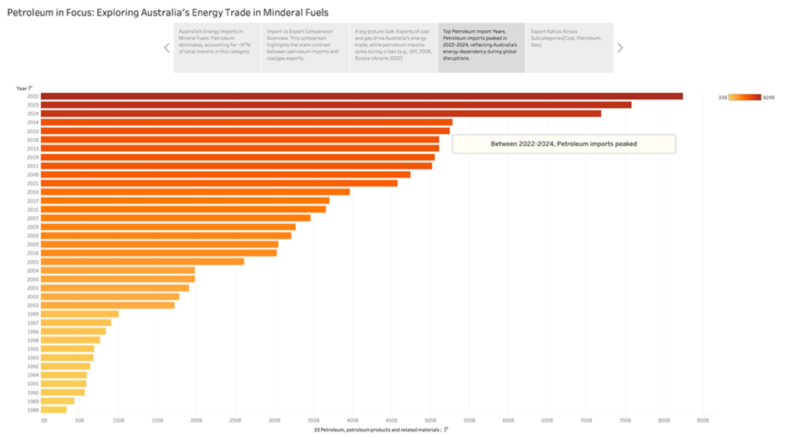
– **Insight**

  - **Petroleum imports are highly volatile, spiking during global crises (GFC 2008, COVID-19, Russia-Ukraine 2022).**
  - **Coal and gas exports are more stable, showing consistent growth.**
  - **Australia's energy trade shows a heavy reliance on petroleum imports, unlike coal/gas.**

– **Key takeaway**

Australia's export strength lies in **coal and gas**, but **petroleum imports** make the country vulnerable during global events.

- **Slide 4:** Yearly Petroleum Import Ranking



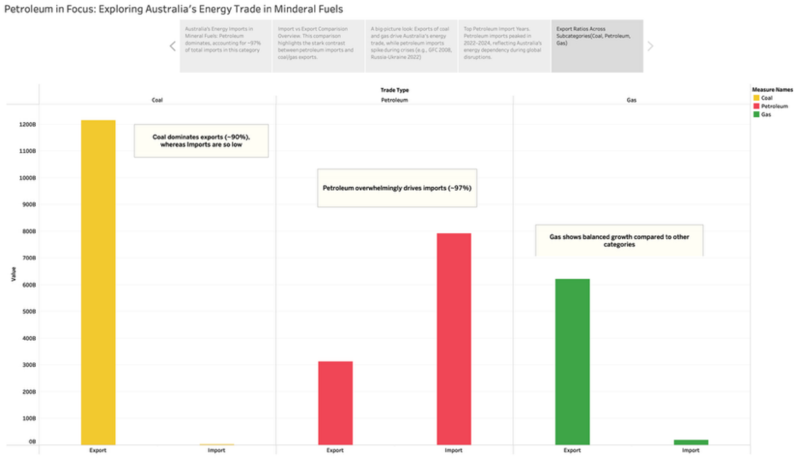
<Figure 18: Top Years in Petroleum Import (1988–2024)>

– **Insight**

  - 2022–2024 mark the highest petroleum import years.
  - Clear upward trend from early 2000s onwards, accelerating post-2008 GFC and post-2020.

– **Key takeaway**: Australia's petroleum import dependency has increased over time, peaking during recent global energy crises.

- **Slide 5: Export vs Import Subcategory Ratios**



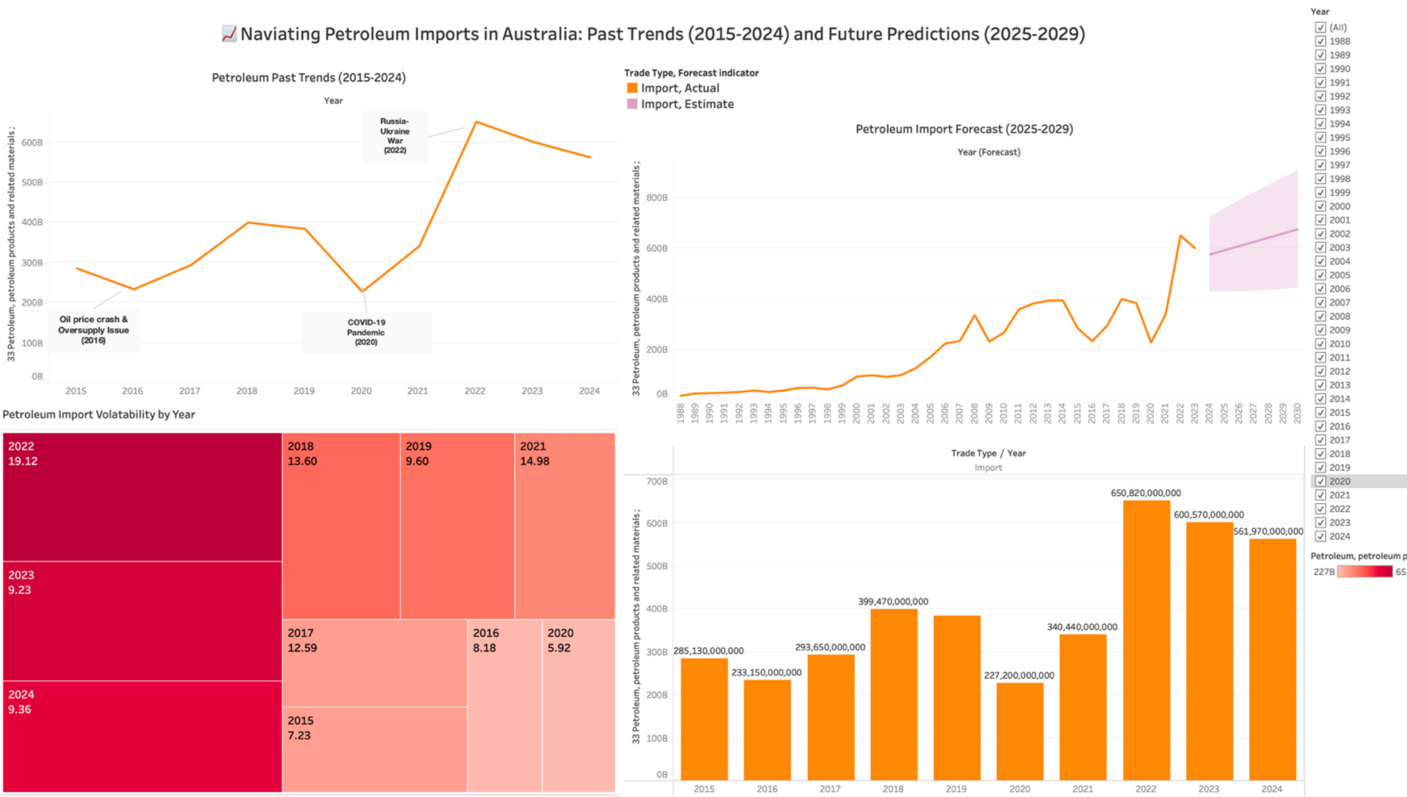
<Figure 19: Export & Import Ratios in Mineral Fuels (Coal, Petroleum, Gas)>

– **Insight**

  - **Export**: Coal dominates (~90%), gas is moderate, petroleum is low.
  - **Import**: Petroleum dominates (~97%), gas is minimal, coal is almost negligible.

– **Key takeaway**: Australia is an energy exporter (coal/gas) but **heavily reliant on petroleum imports**, highlighting an import-export imbalance.

6. Product Analysis and Forecast (2015-2024, 2025-2029)



<Figure 20Product Analysis and Forecast of Petroleum Imports in Australia (2015–2024, 2025–2029)>

6.1 Past Trends Analysis (2015-2024)

Between 2015 and 2024, Australia’s petroleum imports exhibited sharp fluctuations, with significant spikes during major global crises, including the Oil Price Crash (2016), the COVID-19 Pandemic (2020), and the Russia-Ukraine War (2022). These events caused volatile shifts in import volumes, with the peak occurring in 2022. The data highlights Australia’s heavy dependence on external petroleum sources and its exposure to global market disruptions.

6.2 Gain or Loss Interpretation

Australia’s petroleum import trends reveal a net increase over the 10-year period, mainly driven by rising demand and global supply challenges. The highest gains were observed in 2022, reflecting the impacts of the Russia-Ukraine War, while temporary declines were seen in 2020 during the COVID-19 pandemic. Overall, the pattern suggests a structural dependency on petroleum imports rather than a stable trade advantage, indicating potential vulnerabilities in Australia's energy security.

6.3 Forecast Predictions (2025–2029)

Future projections indicate that petroleum imports will remain high, although a gradual decline is expected post-2024. This trend reflects anticipated global market adjustments and efforts to stabilise energy dependency. However, uncertainty remains due to ongoing geopolitical tensions and potential supply chain disruptions. The forecast was developed using a 95% confidence interval, ensuring reasonable accuracy within the predicted range, though external shocks could still cause deviations.

6.4 Future Trade Recommendations

To enhance energy security and reduce vulnerability to external shocks, Australia should diversify its energy sources beyond petroleum. Strengthening domestic supply chains, increasing investment in renewable energy technologies, and promoting local production of alternative fuels are crucial strategies. Implementing these measures could reduce dependency on volatile petroleum imports, enhance resilience to future market crises, and support Australia’s long-term energy and trade sustainability.

7. Conclusion

This analysis explored Australia's energy trade within the *Mineral Fuels, Lubricants, and Related Materials* category, focusing on petroleum as the dominant subcategory. The designed graphs highlighted significant trends: petroleum imports surged during global crises such as the Global Financial Crisis (2008), Oil Price Crash (2014–2016), COVID-19 Pandemic (2020), and the Russia-Ukraine War (2022). These events caused sharp volatility in petroleum imports, whereas coal and gas exports remained relatively stable.

By comparing petroleum to its main category and subcategories (coal and gas), it is evident that petroleum overwhelmingly drives Australia’s import dependency, accounting for over 97% of energy imports in this category, while coal and gas are key export drivers. The dashboard and storyboard effectively visualised these patterns, offering clear insights into Australia's energy trade dynamics over the past 10 years and forecast trends for the next 5 years (2025–2029), developed using a 95% confidence interval for projections.

Based on the analysis, it is recommended that Australia diversify energy sources, reduce over-reliance on petroleum imports, and strengthen domestic production of alternative energy, such as renewables, to mitigate external risks. The dashboard and storyboard provided a clear, interactive, and data-driven presentation, supporting data-backed decision-making for future trade strategies.